Group 2
BSIT-3B

Discrete Mathematics Final Requirement

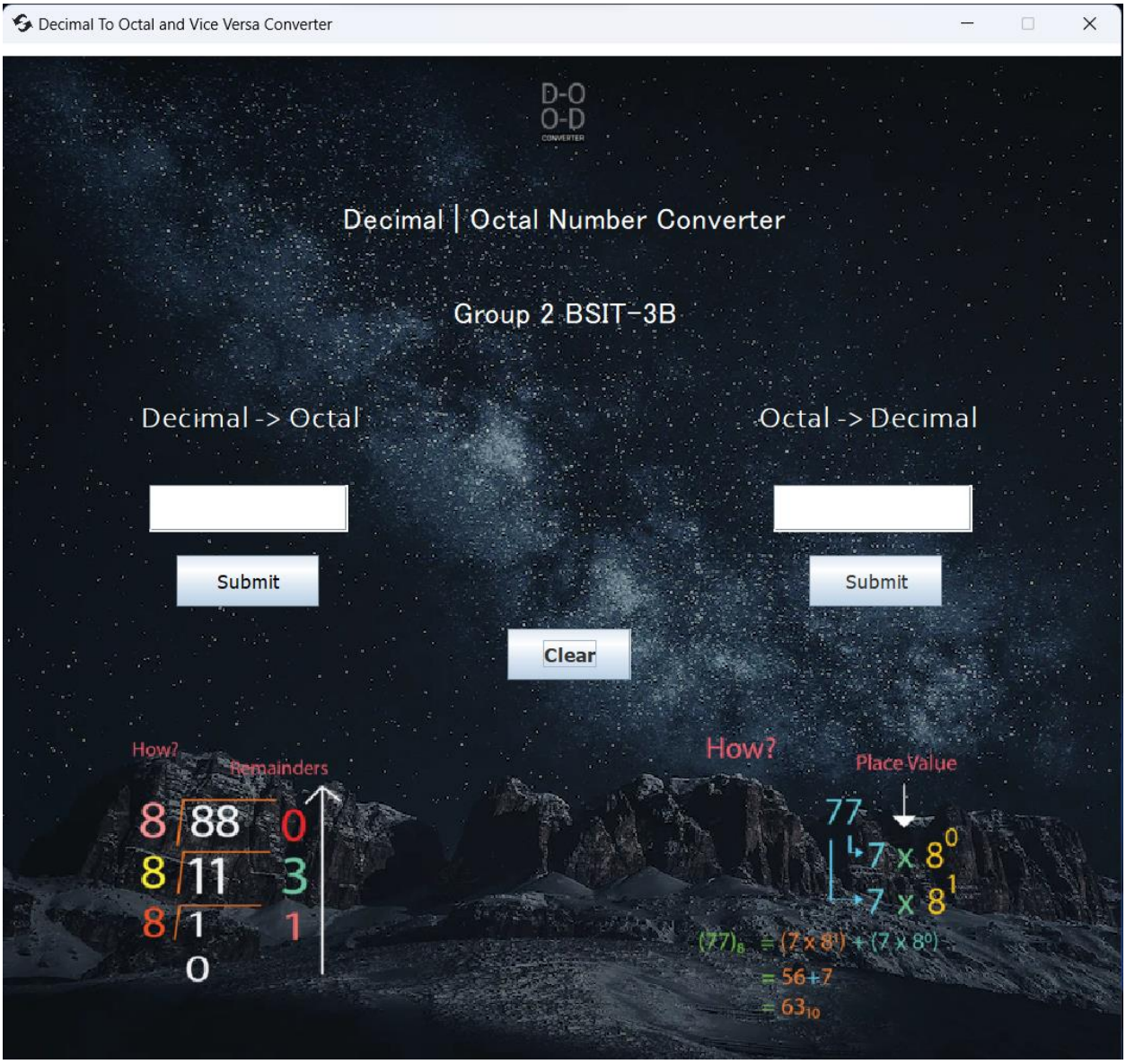
Application Title

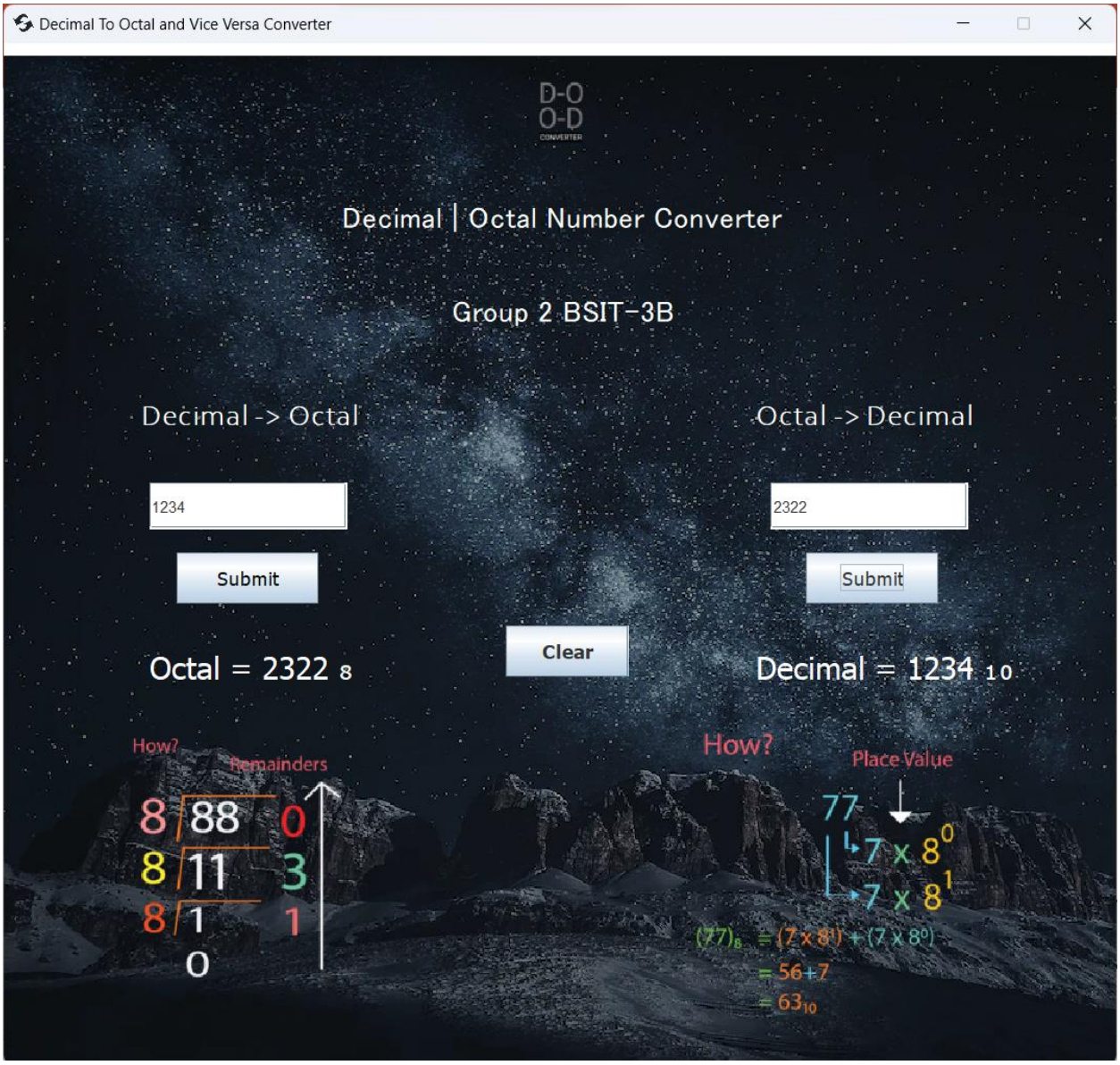
Octal Decimal Vice Versa Converter

Members:

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Raymundo, Vyete
Santiago, Carl Daniel
Sugui, Christine

GUI


Decimal To Octal and Vice Versa Converter
D-O
O-D
CONVERTER
Decimal | Octal Number Converter
Group 2 BSIT-3B
Decimal -> Octal
Octal -> Decimal
Submit
Submit
Clear
How?
Remainders
8 | 88 0
8 | 11 3
8 | 1 1
0
How?
Place Value
77
7 x 8^0
7 x 8^1
(77)8 = (7 x 8^1) + (7 x 8^0)
= 56+7
= 63₁₀


Decimal To Octal and Vice Versa Converter
D-O
O-D
CONVERTER
Decimal | Octal Number Converter
Group 2 BSIT-3B
Decimal -> Octal
1234
Submit
Octal = 2322 8
Clear
Octal -> Decimal
2322
Submit
Decimal = 1234 10
How?
Remainders
8 88 0
8 11 3
8 1 1
0
How?
Place Value
77
7 x 8^0
7 x 8^1
(77)8 = (7 x 8^1) + (7 x 8^0)
= 56+7
= 63₁₀

Program and Software Used



An integrated development environment called Eclipse
is employed in computer programming. An integrated development
environment that is used in computer programming,
and it is the second-most popular IDE for Java development.
.



Java is a high-level, class-based, object-oriented programming language.
general-purpose programming language. It is used to develop desktop and
mobile applications, big data processing, embedded systems etc



Launch4j is a cross-platform tool for wrapping
Java applications distributed as jars in lightweight Windows native executables.

Source Code

```
import java.awt.EventQueue;
/* Group 2
 * Lead by: Carl Daniel Santiago
 *
 * Members
 * Izellah Cillo
 * Kenneth Galang
 * Yvete Raymundo
 * Christine Suqui
 *
 *
 */
import javax.swing.JFrame;
import javax.swing.JPanel;
import javax.swing.border.EmptyBorder;
import javax.swing.GroupLayout;
import javax.swing.GroupLayout.Alignment;
import javax.swing.ImageIcon;
import javax.swing.JLabel;
import javax.swing.SwingConstants;
import java.awt.Font;
import java.awt.Image;

import javax.swing.LayoutStyle.ComponentPlacement;
import javax.swing.JTextField;
import javax.swing.JButton;
import java.awt.event.ActionListener;
import java.awt.event.ActionEvent;
```

```
import java.awt.EventQueue;

public class ConverterOctDec extends JFrame {

    private JPanel contentPane;
    private JTextField textField;
    private JTextField textField2;

    /**
     * Launch the application.
     */
    public static void main(String[] args) {
        EventQueue.invokeLater(new Runnable() {
            public void run() {
                try {
                    ConverterOctDec frame = new ConverterOctDec();
                    frame.setVisible(true);
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }
        });
    }


    /**
     * Create the frame.
     */
    public ConverterOctDec() {
        setResizable(false);
        setIconImage(Toolkit.getDefaultToolkit().getImage("C:\\Users\\CarlS\\Desktop\\Assets Converter OctDec\\icon-convert-0.jpg"));
        setTitle("Decimal To Octal and Vice Versa Converter\r\n");
        setFont(new Font("SimSun-ExtB", Font.PLAIN, 12));
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setBounds(100, 100, 869, 821);
        contentPane = new JPanel();
```

```
67          setBounds(100, 100, 869, 821);
68          contentPane = new JPanel();
69          contentPane.setBackground(Color.WHITE);
70          contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));
71
72          setContentPane(contentPane);
73
74          JLabel lblNewLabel = new JLabel("Decimal | Octal Number Converter");
75          lblNewLabel.setForeground(new Color(253, 255, 255));
76          lblNewLabel.setBounds(240, 101, 378, 66);
77          lblNewLabel.setBackground(Color.WHITE);
78          lblNewLabel.setFont(new Font("MS PGothic", Font.PLAIN, 23));
79          lblNewLabel.setHorizontalAlignment(SwingConstants.CENTER);
80
81          JLabel lblGroupBsitb = new JLabel("Group 2 BSIT-3B");
82          lblGroupBsitb.setForeground(new Color(253, 255, 255));
83          lblGroupBsitb.setBounds(299, 177, 261, 59);
84          lblGroupBsitb.setHorizontalAlignment(SwingConstants.CENTER);
85          lblGroupBsitb.setFont(new Font("MS PGothic", Font.PLAIN, 23));
86
87
88          textField = new JTextField();
89          textField.setBounds(112, 337, 152, 36);
90          textField.addKeyListener(new KeyAdapter() {
91
92              @Override
93
94
95
96              public void keyTyped(KeyEvent e) {
97
98                  char num = e.getKeyChar();
99
100                 if (!Character.isDigit(num)) {
101                     e.consume();
102                 }
```

```
94
95
96              public void keyTyped(KeyEvent e) {
97
98                  char num = e.getKeyChar();
99
100                 if (!Character.isDigit(num)) {
101                     e.consume();
102                 }
103
104             }
105         });
106         textField.setColumns(10);
107
108         JLabel lblDecimalOctal = new JLabel("Decimal -> Octal");
109         lblDecimalOctal.setForeground(new Color(255, 255, 255));
110         lblDecimalOctal.setBounds(50, 260, 278, 59);
111         lblDecimalOctal.setHorizontalAlignment(SwingConstants.CENTER);
112         lblDecimalOctal.setFont(new Font("Corbel Light", Font.BOLD, 23));
113
114         JLabel lblOctalOctal = new JLabel("Octal -> Decimal");
115         lblOctalOctal.setForeground(new Color(255, 255, 255));
116         lblOctalOctal.setBounds(531, 260, 261, 59);
117         lblOctalOctal.setHorizontalAlignment(SwingConstants.CENTER);
118         lblOctalOctal.setFont(new Font("Corbel Light", Font.BOLD, 23));
119
120         textField2 = new JTextField();
121         textField2.setBounds(589, 337, 152, 36);
122         textField2.addKeyListener(new KeyAdapter() {
123             @Override
124             public void keyTyped(KeyEvent e) {
125
126                 char num2 = e.getKeyChar();
127
128                 if (!Character.isDigit(num2)) {
129                     e.consume();
```

```
        textField2.setBounds(589, 337, 152, 36);
        textField2.addKeyListener(new KeyAdapter() {
            @Override
            public void keyTyped(KeyEvent e) {

                char num2 = e.getKeyChar();

                if (!Character.isDigit(num2)) {
                    e.consume();
                }
            }
        });
        textField2.setColumns(10);

        JLabel Label_DecToOct = new JLabel("");
        Label_DecToOct.setForeground(new Color(255, 255, 255));
        Label_DecToOct.setBounds(72, 460, 237, 36);
        Label_DecToOct.setHorizontalAlignment(SwingConstants.CENTER);
        Label_DecToOct.setFont(new Font("Tahoma", Font.PLAIN, 24));

        // Show for Decimal to Octal

        JLabel Label_OctToDec = new JLabel("");
        Label_OctToDec.setForeground(new Color(255, 255, 255));
        Label_OctToDec.setBounds(562, 460, 230, 36);
        Label_OctToDec.setHorizontalAlignment(SwingConstants.CENTER);
        Label_OctToDec.setFont(new Font("Tahoma", Font.PLAIN, 24));

        // Show for Octal to Decimal


        JButton btnNewButtonDecToOct = new JButton("Submit");
        btnNewButtonDecToOct.setForeground(new Color(0, 0, 0));
        btnNewButtonDecToOct.setBounds(133, 391, 109, 39);
        btnNewButtonDecToOct.addActionListener(new ActionListener() {

```

```

        JButton btnNewButtonDecToOct = new JButton("Submit");
        btnNewButtonDecToOct.setForeground(new Color(0, 0, 0));
        btnNewButtonDecToOct.setBounds(133, 391, 109, 39);
        btnNewButtonDecToOct.addActionListener(new ActionListener() {


            public void actionPerformed(ActionEvent e) {


                String a = textField.getText();
                int b=Integer.parseInt(a);
                Label_DecToOct.setText("Octal = "+Integer.toOctalString(b)+" ₈");


            }
        });

        // BTN for Decimal to Octal

        btnNewButtonDecToOct.setFont(new Font("Tahoma", Font.PLAIN, 15));


        JButton btnNewButtonOctToDec = new JButton("Submit");
        btnNewButtonOctToDec.setBounds(616, 391, 102, 39);
        btnNewButtonOctToDec.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {

                String alpha = textField2.getText();
                int decimal=Integer.parseInt(alpha,8);
                Label_OctToDec.setText("Decimal = "+decimal+" ₁₀");


            }
        });

```

```
            }
        });

        // BTN for Octal to Decimal
        btnNewButtonOctToDec.setFont(new Font("Tahoma", Font.PLAIN, 15));

        JButton btnNewButton = new JButton("Clear");
        btnNewButton.setBounds(386, 447, 94, 39);
        btnNewButton.setFont(new Font("Tahoma", Font.BOLD, 15));
        btnNewButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {

                // This Button Reset Everything to no value

                Label_DecToOct.setText("");
                Label_OctToDec.setText("");
                textField.setText("");
                textField2.setText("");
            }
        });

        JLabel logoPlace = new JLabel("");
        logoPlace.setBackground(new Color(253, 255, 255));
        logoPlace.setBounds(389, 15, 80, 80);
        logoPlace.setIcon(new ImageIcon("C:\\Users\\CarlS\\Desktop\\Assets Converter OctDec\\logo lastyyy.png"));

        JLabel explanationDecOct = new JLabel("");
        explanationDecOct.setBounds(99, 505, 184, 245);
        explanationDecOct.setIcon(new ImageIcon("C:\\Users\\CarlS\\Desktop\\Assets Converter OctDec\\DoO5.png"));

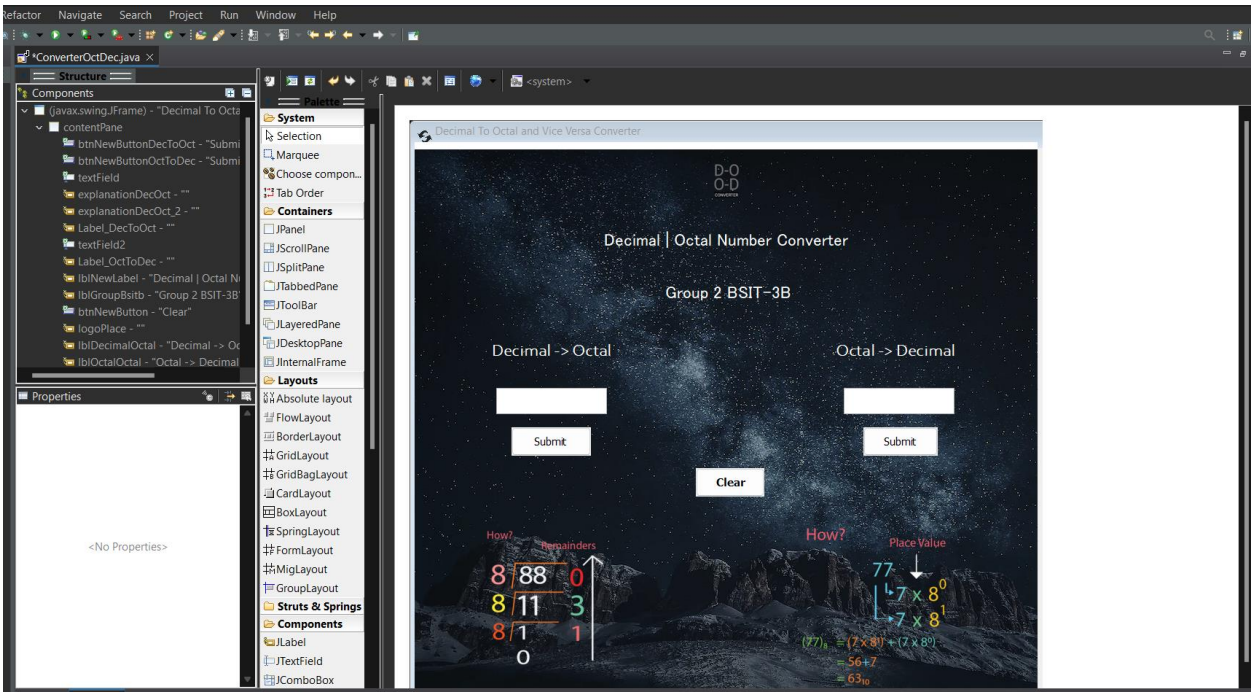
        JLabel explanationDecOct_2 = new JLabel("");
        explanationDecOct_2.setBounds(531, 523, 261, 227);
        explanationDecOct_2.setIcon(new ImageIcon("C:\\Users\\CarlS\\Desktop\\Assets Converter OctDec\\ytess8909.png"));
        contentPane.setLayout(null);
        contentPane.add(btnNewButtonDecToOct);
        contentPane.add(btnNewButtonOctToDec);
```

```
        JLabel logoPlace = new JLabel("");
        logoPlace.setBackground(new Color(253, 255, 255));
        logoPlace.setBounds(389, 15, 80, 80);
        logoPlace.setIcon(new ImageIcon("C:\\Users\\CarlS\\Desktop\\Assets Converter OctDec\\logo lastyyy.png"));

        JLabel explanationDecOct = new JLabel("");
        explanationDecOct.setBounds(99, 505, 184, 245);
        explanationDecOct.setIcon(new ImageIcon("C:\\Users\\CarlS\\Desktop\\Assets Converter OctDec\\DoO5.png"));

        JLabel explanationDecOct_2 = new JLabel("");
        explanationDecOct_2.setBounds(531, 523, 261, 227);
        explanationDecOct_2.setIcon(new ImageIcon("C:\\Users\\CarlS\\Desktop\\Assets Converter OctDec\\ytess8909.png"));
        contentPane.setLayout(null);
        contentPane.add(btnNewButtonDecToOct);
        contentPane.add(btnNewButtonOctToDec);
        contentPane.add(textField);
        contentPane.add(explanationDecOct);
        contentPane.add(explanationDecOct_2);
        contentPane.add(Label_DecToOct);
        contentPane.add(textField2);
        contentPane.add(Label_OctToDec);
        contentPane.add(lblNewLabel);
        contentPane.add(lblGroupBsitb);
        contentPane.add(btnNewButton);
        contentPane.add(logoPlace);
        contentPane.add(lblDecimalOctal);
        contentPane.add(lblOctalOctal);

        JLabel lblNewLabel_1 = new JLabel("");
        lblNewLabel_1.setIcon(new ImageIcon("C:\\Users\\CarlS\\Desktop\\Assets Converter OctDec\\background.jpg"));
        lblNewLabel_1.setBounds(0, 0, 855, 801);
        contentPane.add(lblNewLabel_1);
    }
}
```


Refactor Navigate Search Project Run Window Help
*ConverterOctDec.java
Structure
Components
(javax.swing.JFrame) - "Decimal To Octa
contentPane
btnNewButtonDecToOct - "Submi
btnNewButtonOctToDec - "Submi
textField
explanationDecOct - ""
explanationDecOct_2 - ""
Label_DecToOct - ""
textField2
Label_OctToDec - ""
lblNewLabel - "Decimal | Octal N
lblGroupBsitb - "Group 2 BSIT-3B
btnNewButton - "Clear"
logoPlace - ""
lblDecimalOctal - "Decimal -> Oc
lblOctalOctal - "Octal -> Decimal
Properties
<No Properties>
Palette
System
Selection
Marquee
Choose compon...
Tab Order
Containers
JPanel
JScrollPane
JSplitPane
JTabbedPane
JToolBar
JLayeredPane
JDesktopPane
JInternalFrame
Layouts
Absolute layout
FlowLayout
BorderLayout
GridLayout
GridBagLayout
CardLayout
BoxLayout
SpringLayout
FormLayout
MigLayout
GroupLayout
Struts & Springs
Components
JLabel
JTextField
JComboBox
<system>
Decimal To Octal and Vice Versa Converter
D-O
O-D
Decimal | Octal Number Converter
Group 2 BSIT-3B
Decimal -> Octal
Octal -> Decimal
Submit
Submit
Clear
How?
Remainders
8 88 0
8 11 3
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7 x 8^0
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(77)8 = (7 x 8^1) + (7 x 8^0)
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